

Fahim Morshed

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RESEARCH INTEREST

Applied Machine Learning, Computer Security, Natural Language Processing

EDUCATION AND CERTIFICATIONS

Bangladesh University of Engineering and Technology (BUET)

Dhaka

Bachelor of Science in Computer Science and Engineering

February 2017 - May 2022

- **CGPA: 3.47/4.00**
- **CGPA of last 2 years: 3.77/4.00**

RESEARCH EXPERIENCE

CORONOSIS

May 2020 - January 2021

Supervisor: [Dr. A. B. M. Alim Al Islam](#) (BUET)

Status: Published in **ACM COMPASS-2022** [[PDF](#)] [[Video Presentation](#)]

- In this project, we developed an integrated dashboard for the COVID-19 pandemic where we performed several predictions (next-day predictions, when would we keep corona under control, country-wise affected and dead, etc.) using epidemiology models.
- We factored in lockdown, social distancing, and contact structures for developing the epidemiology models. These analyses were demonstrated through interactive charts, graphs, and maps.
- I conducted front-end implementation of maps and charts to visualize multiple factors, assisted in data collection, assisted in building the model and took part in writing the paper draft.

Downstream Tasks in Bangla NLP

January 2021 - Present

Supervisor: [Dr. Anindya Iqbal](#) (BUET)

- Explored two downstream tasks in a low resource language such as Bangla as my undergraduate thesis leveraging transfer learning of large language models namely GPT-2 and MT-5.
- We attempted Bangla text style transfer (from Sadhu to Colito and vice versa) and achieved satisfactory result. To fine-tune MT-5 and GPT-2 we synthesized a dataset of parallel Sadhu-Colito sentences by a multi-step process involving rule based method, hand curating and model output. Additionally, we proposed an evaluation scheme consisting of both automatic and manual effort to benchmark our results. We are almost finished with writing the manuscript and we will try for publication.
- We explored paraphrasing ability of Bangla pretrained GPT-2 and MT-5 with the help of a large Bangla newspaper corpus.
- I conducted data collection, data pre-processing, finding optimal hyperparameters for models, writing scripts for evaluation and assisted in many other aspects collaborating with my thesis partner.

A Bimodal analysis on sentiment changes in Twitter during the Pandemic

May 2021 - Present

Supervisor: [Dr. A. B. M. Alim Al Islam](#) (BUET)

Status: Under Review in **CHI 2023**

- We investigated changes in human sentiment over two years on Twitter during the COVID-19 pandemic by incorporating bimodal analyses of tweets and images.
- Collected and annotated tweets over the internet for the timeline of 2019-2020 and performed exploratory analysis on the data that portrays the change in sentiment in social media for a large timeline during the pandemic.
- I worked on manual tweet image caption labeling and experimented with image captioning models to generate tweet captions.

PROFESSIONAL EXPERIENCE

Software Developer

September 2022 - Present

Employer: [IQVIA](#)

- Working in the Auth-Services team of this Fortune 500 company where the team's responsibility is to maintain a variety of user activities including authentication and authorization.
- Collaborating with team members from different parts of the world.

- Learning highly advanced enterprise frameworks, technologies and international standard work culture while carrying out my assigned tasks.
- Implementing and improving features of Angular web application and ASP.NET back-end service.
- Writing modular, scalable and reusable codes along with necessary tests.

Software Developer

April 2022 - October 2022

Employer: [Department of CSE, BUET](#)

- Been a part of large government project IEIMS of BANBEIS working in the Result Processing module.
- Replaced manual efforts in the exam evaluation pipeline by automated solutions.
- Wrote codes for React reusable components, Laravel eloquent queries and logics in API, MSSQL DB stored procedures.
- Presented progress to government stakeholders.

Junior Software Engineer (Part-time)

September 2020 - May 2021

Employer: [Sohopathi](#)

- Developed and maintained React-native mobile applications.
- Integrated APIs in react-native front-end and leveraged state management of redux-toolkit.
- Designed responsive web pages to properly show across multiple devices, e.g. mobile, tab, desktop.
- Extensively worked in every stages of MERN stack.

PROJECTS

Logistic Regression and Adaboost <i>Machine Learning Project</i>	2022
• Implemented Logistic Regression and Adaboost from scratch.	
Kemon Achen <i>Software Development Project</i>	2022
• A mental healthcare mobile app using react native as front end and node as backend.	
Attacks and AES <i>Computer Security Projects</i>	2021
• Three fundamental Computer Security projects: ARP cache poisoning + Man in the middle attack, Cross site scripting (XSS) attack and Advanced Encryption Standard (AES) algorithm.	
Rikshaw Fare Meter <i>Micro-controller Project</i>	2019
• Built a fare calculating device with multiple sensors for a local vehicle called Rikshaw.	

AWARDS

- Dean's List (Level-4): Bangladesh University of Engineering and Technology (BUET)
- Board Merit Scholarship: Government of Bangladesh (Year: 2012, 2014, 2016)
- Honourable Mention (3rd Position) at a Hackathon organized by BRAC University

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, PHP, SQL, C#, Kotlin, Bash, YAML

Database: PostgreSQL, MongoDB, MS Sql Server

Frameworks: React, Angular, Express.js, ASP.NET, React-native, Native Android **Tools:** LASER, LaBSE, FAISS, Huggingface

REFERENCE

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